



Teacher answer keys

Keep separate from student practice pages.

Accept equivalent unreduced answers and valid nonleast common denominators. Numerical correctness, written method, and help provided are separate observations.

Baseline: show your thinking

1. $\frac{1}{2} = \frac{3}{6}$; $\frac{1}{3} = \frac{2}{6}$; $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$
2. $\frac{1}{4} = \frac{3}{12}$; $\frac{2}{3} = \frac{8}{12}$; $\frac{3}{12} + \frac{8}{12} = \frac{11}{12}$
3. $\frac{2}{5} = \frac{4}{10}$; $\frac{4}{10} + \frac{1}{10} = \frac{5}{10}$; $\frac{5}{10} = \frac{1}{2}$
4. $\frac{1}{4} = \frac{2}{8}$; $\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$; The class uses $\frac{5}{8}$ meter altogether.

Targeted: equal parts before adding

1. $\frac{1}{2} = \frac{2}{4}$; $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$. The strips must represent the same-sized whole.
2. $\frac{1}{5} = \frac{2}{10}$; $\frac{2}{10} + \frac{1}{10} = \frac{3}{10}$.
3. Halves and fourths are different-sized parts. Rename $\frac{1}{2}$ as $\frac{2}{4}$ and add $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$. The answer $\frac{2}{6}$ is smaller than $\frac{1}{2}$, so it cannot be the sum of these positive addends.

Independent entry check

1. $\frac{1}{2} = \frac{2}{4}$; $\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$.
2. $\frac{1}{2} = \frac{3}{6}$; $\frac{1}{6} + \frac{3}{6} = \frac{4}{6} = \frac{2}{3}$.

Independent application

1. $\frac{1}{4} = \frac{5}{20}$; $\frac{1}{5} = \frac{4}{20}$; total $\frac{9}{20}$ meter.
2. $\frac{1}{5} = \frac{2}{10}$; $\frac{1}{2} = \frac{5}{10}$; total $\frac{7}{10}$ meter.



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Extension: two methods, one value

1. $5/20 + 8/20 = 13/20$; $10/40 + 16/40 = 26/40 = 13/20$.
2. Examples: $1/2 + 1/4 = 3/4$; $1/3 + 5/12 = 4/12 + 5/12 = 9/12 = 3/4$. Accept other valid pairs.
3. An explanation should connect denominator to part size, rename unlike units into the same unit, and retain that denominator when adding numerators.

Lesson exit ticket

1. $1/4 = 3/12$; $1/6 = 2/12$; total $5/12$.
2. Accept an explanation that the fractions need the same-sized parts/common unit before their counts can be added.

Follow-up: a fresh fraction check

1. $1/3 = 4/12$; $1/4 = 3/12$; $4/12 + 3/12 = 7/12$
2. $1/3 = 2/6$; $1/6 + 2/6 = 3/6$; $3/6 = 1/2$; Jules uses $1/2$ meter altogether.

Fraction practice

1. $1/4 = 2/8$; $2/8 + 1/8 = 3/8$
2. $1/3 = 5/15$; $1/5 = 3/15$; $5/15 + 3/15 = 8/15$
3. $1/3 = 3/9$; $2/9 + 3/9 = 5/9$; Both addends now count ninths.

Word problems

1. $1/5 = 2/10$; $2/10 + 3/10 = 5/10 = 1/2$ meter
2. $2/5 = 6/15$; $1/3 = 5/15$; $6/15 + 5/15 = 11/15$ meter
3. $1/4 = 5/20$; $1/10 = 2/20$; $5/20 + 2/20 = 7/20$ meter



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Independent check

1. $\frac{1}{2} = \frac{5}{10}$; $\frac{5}{10} + \frac{3}{10} = \frac{8}{10} = \frac{4}{5}$
2. $\frac{1}{5} = \frac{3}{15}$; $\frac{2}{3} = \frac{10}{15}$; $\frac{3}{15} + \frac{10}{15} = \frac{13}{15}$
3. $\frac{1}{4} = \frac{3}{12}$; $\frac{3}{12} + \frac{1}{12} = \frac{4}{12} = \frac{1}{3}$ meter

Support and interpretation

Record hints and worked examples as support. A blank response means there is not enough evidence yet. The source worksheets and this answer key are synthetic fictional examples; teacher review remains required.